

The Principle of Least Sufficient Reaction

A Formal Statement for AHR-v3

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Abstract

The classical Principle of Least Action characterizes the realized path of a system as the one for which the action is stationary [3, 5, 9]. It does not, however, prescribe how a supervisory layer should respond to deviations from an expected behavioral path. AHR-v3 supplies that second half by coupling a least-action substrate to a least-sufficient-reaction policy layer through a native time-indexed identity (tID). This document formalizes the resulting principle, its admissible reaction set, its coupling condition, and its relationship to bounded interference and the Amdahl bound [2]. The principle is presented as a proposed control abstraction, not as an established hardware capability.

1 Motivation

The classical Principle of Least Action states that the realized path of a system is the one for which the action is stationary [3, 5, 9]:

$$\delta S = \delta \int_{t_1}^{t_2} L(q, \dot{q}, t) dt = 0. \quad (1)$$

This principle describes *what the system does*. It does not, by itself, prescribe *how a supervisory layer should respond* to deviations from an expected behavioral path. The AHR-v3 proposal adds that second half: the least action must generate a *least sufficient reaction*, with both sides linked by time and a native tID. The present formalization is close in spirit to the supervisory formulations in [7, 8] and to the recomputing-environment model of [1].

2 The Native tID as the Shared Time Base

Let the timing quantum be Δt and let t_0 be the epoch. The native tID is the time-indexed identity of an execution segment:

$$\text{tID}(t) = \left\lfloor \frac{t - t_0}{\Delta t} \right\rfloor. \quad (2)$$

Each tID defines a half-open window

$$W_k = [t_0 + k \Delta t, t_0 + (k + 1) \Delta t], \quad k = \text{tID}(t). \quad (3)$$

Every action sample and every reaction decision is labeled by the same k , so that action and reaction are *temporally identified* rather than merely correlated by wall-clock time:

$$E_i = (\text{tID}_i, \text{node}_i, \text{event}_i, \text{severity}_i). \quad (4)$$

3 Behavioral Node and Deviation Vector

A behavioral node is

$$n_i = (\mathbf{x}_i, \mathbf{r}_i, \mathbf{t}_i, \mathbf{q}_i), \quad (5)$$

where $\mathbf{x}_i$ is execution state, $\mathbf{r}_i$ resource allocation, $\mathbf{t}_i$ timing state, and $\mathbf{q}_i$ behavioral or policy state [8]. The behavioral vector is the image of a mapping into a precomputed hierarchy H :

$$\mathbf{v}_i(t) = \Phi(n_i(t), \mathcal{N}, \mathbf{c}_i(t)), \quad (6)$$

with $\mathbf{c}_i(t)$ the current context, resource, and timing information.

The *angular severity deviation* relative to a permitted reference behavior $\mathbf{v}_{\text{ref}}(t)$ is

$$\theta_i(t) = \arccos\left(\frac{\mathbf{v}_i(t) \cdot \mathbf{v}_{\text{ref}}(t)}{\|\mathbf{v}_i(t)\| \|\mathbf{v}_{\text{ref}}(t)\|}\right). \quad (7)$$

The scalar severity score combines angular, resource, and latency deviation:

$$S_i(t) = \lambda_\theta \theta_i(t) + \lambda_r R_i(t) + \lambda_\ell L_i(t), \quad \lambda_\theta, \lambda_r, \lambda_\ell > 0. \quad (8)$$

4 Action Functional at tID Granularity

For node i during window W_k , define the action

$$\mathcal{S}_i^{(k)} = \int_{W_k} L_i(\mathbf{x}_i, \dot{\mathbf{x}}_i, t) dt, \quad (9)$$

with Lagrangian L_i [5]. The classical stationarity condition $\delta \mathcal{S}_i^{(k)} = 0$ describes the node's realized path within the window. AHR-v3 treats this as the *action side* of the principle.

4.5 Behavioral Envelopes and Sandbox Classification

The reference vector $\mathbf{v}_{\text{ref}}(t)$ should not be a single global point. Instead, each sandbox i is assigned a behavioral class $h_i \in H$, and deviation is measured against the class-specific envelope $E(h_i) \subset H$.

The behavioral severity is

$$\theta_i(t) = \arccos\left(\frac{\mathbf{v}_i(t) \cdot \mathbf{v}_{h_i}(t)}{\|\mathbf{v}_i(t)\| \|\mathbf{v}_{h_i}(t)\|}\right), \quad (10)$$

where $\mathbf{v}_{h_i}(t)$ is the centroid of the envelope $E(h_i)$.

A sandbox is *inside its envelope* when

$$\theta_i(t) \leq \theta_{\text{envelope}}(h_i). \quad (11)$$

Capacity pressure is a separate signal:

$$P_i(t) = \lambda_r R_i(t) + \lambda_\ell L_i(t). \quad (12)$$

Behavioral severity feeds Principle 1 escalation. Capacity pressure feeds the scheduler cost function.

This separation prevents false positives from high-intensity but authorized workloads.

5 Admissible Reaction Set

Let the AtID of thread/node j be

$$\text{AtID}_j = (\text{tenant}_j, \text{sandbox}_j, \text{capability}_j, \text{tID}_j). \quad (13)$$

Access is gated by the capability graph Γ [6]:

$$\text{allow}(j, a) = \begin{cases} 1, & \text{if } a \in \Gamma(\text{AtID}_j), \\ 0, & \text{otherwise.} \end{cases} \quad (14)$$

The set of reactions admissible at time t for node i is therefore

$$\mathcal{A}_i(t) = \{a \in \mathcal{A} \mid a \in \Gamma(\text{AtID}_i(t))\}, \quad (15)$$

where $\mathcal{A}$ is the universe of supervisory reactions (accept, delay, isolate, roll back, reject, throttle, migrate, and so on)[8, 1].¹

6 The Principle of Least Sufficient Reaction

Principle 1 (Least Sufficient Reaction). Given a severity threshold $S_{\text{threshold}}$ and an admissible set $\mathcal{A}_i(t)$, the AHR-v3 reaction is the element of least magnitude that keeps the node within policy:

$$a_i^*(t) = \arg \min_{a \in \mathcal{A}_i(t)} \{\|a\| \mid S_i(t) \leq S_{\text{threshold}}\}. \quad (16)$$

If no admissible reaction satisfies the bound, the reaction escalates to the next admissible class until either the bound is met or the node is isolated.

Equivalently, in constrained form,

$$\begin{aligned} a_i^*(t) &= \arg \min_{a \in \mathcal{A}_i(t)} \|a\| \\ \text{s.t. } S_i(t) &\leq S_{\text{threshold}}, \\ a &\in \Gamma(\text{AtID}_i(t)), \\ \text{tID}(a) &= \text{tID}(S_i(t)). \end{aligned} \quad (17)$$

7 Coupling Least Action to Least Reaction

The new proposal is not either half alone, but their coupling through the native tID:

$$\boxed{\text{least action} \wedge \text{least sufficient reaction linked by native tID}} \quad (18)$$

Formally, for each window W_k the pair

$$(\mathcal{S}_i^{(k)}, a_i^{*,(k)}) \quad (19)$$

is evaluated at the *same* $k = \text{tID}(t)$, so that the reaction is indexed by the action that produced it. This yields a causal chain that is observationally closed within a single tID:

$$\text{action sample at } k \rightarrow S_i \rightarrow a_i^* \rightarrow \text{same } k. \quad (20)$$

¹The admissible set $\mathcal{A}_i(t)$ is gated by AtID, but the triggering condition is $\theta_i(t) > \theta_{\text{envelope}}(h_i)$, not raw severity. A sandbox can have high severity (high resource usage) without triggering escalation if its behavioral deviation is low. This resolves the ambiguity identified without changing the core mathematics—it just makes the inputs to the mathematics more precise.

8 Recomputing Environment

Because reactions are recomputed per tID window, AHR-v3 does not execute a fixed instruction sequence. Instead it repeatedly evaluates behavior, resources, and timing [1]:

$$\mathcal{R}_v = \{K, \Pi, \Gamma, \Lambda\}, \quad (21)$$

with K knowledge, Π policy, Γ capability graph, and Λ the timing and resource model. Sandboxes submit proposed transitions

$$\Delta_i = F(n_i, \mathcal{R}_v, \mathbf{v}_i), \quad (22)$$

and the supervisory layer returns

$$a_i(t) = \Psi(\Delta_i(t), \mathbf{v}_i(t), \Gamma, \Lambda), \quad (23)$$

which under Principle 1 specializes to $a_i^*(t)$.

9 Bounded Interference as the Design Goal

Least sufficient reaction is not perfect isolation. The goal is bounded interference:

$$P(L_i > L_{\text{target}}) \leq \epsilon, \quad (24)$$

with L_i the observed latency of node i and ϵ the permitted tail probability [4]. The scheduler cost that ranks candidate placements is

$$\text{cost}(i, s) = \alpha L_{i,\text{local}} + \beta L_{i,\text{remote}} + \gamma C_{i,\text{migration}} + \delta J_{i,\text{queue}}. \quad (25)$$

10 Amdahl Bound on the Reaction Path

The serial fraction of the control path bounds the achievable speedup [2]:

$$S_{\text{Amdahl}} = \frac{1}{(1-p) + \frac{p}{s}}, \quad \lim_{s \rightarrow \infty} S_{\text{Amdahl}} = \frac{1}{1-p}. \quad (26)$$

The reaction path itself contributes to the serial fraction. Least sufficient reaction is therefore also a strategy for keeping the serial critical path short: the smallest reaction that satisfies policy is the one that adds the least coordination overhead to the next tID window.

11 Statement of the New Principle

Principle 2 (AHR-v3 Least Policy Reaction). The AHR-v3 supervisory layer shall, for every tID window, select the reaction of least magnitude that is admissible under the node’s AtID and that keeps the severity score within its threshold, with action and reaction sharing the same native tID.

In compact form:

$$a_i^*(t) = \arg \min_{a \in \Gamma(\text{AtID}_i)} \{ \|a\| \mid S_i(t) \leq S_{\text{threshold}}, \text{tID}(a) = \text{tID}(t) \} \quad (27)$$

12 Relationship to Classical Least Action

$$\begin{array}{ll} \text{Classical:} & \delta\mathcal{S} = 0 \quad (\text{single variational statement}) \\ \text{AHR-v3:} & \delta\mathcal{S} = 0 + \min \|a\| \text{ s.t. } S_i \leq S_{\text{threshold}}, \text{ indexed by tID} \end{array} \quad (28)$$

The novelty is not the minimization of action. The novelty is the coupling of a least-action substrate to a least-sufficient-reaction policy layer, with the native tID as the shared time base that makes the coupling observable, enforceable, and reproducible [7, 9].

13 Remark on Status

Remark 1. As with the rest of AHR-v3, this principle is a *proposed control abstraction*. It should not be presented as an established capability of NVIDIA Vera or of any current hardware until an implementation and benchmark demonstrate it [8].

References

- [1] AHR Project. Virtual ROM and the recomputing environment, October 2026. Companion note to AHR-v3.
- [2] Gene M. Amdahl. Validity of the single processor approach to achieving large-scale computing capabilities. *AFIPS Spring Joint Computer Conference*, pages 483–485, 1967.
- [3] Pierre Louis Moreau de Maupertuis. Accord de différentes lois de la nature qui avaient jusqu’ici paru incompatibles. *Mémoires de l’Académie Royale des Sciences*, pages 417–426, 1744.
- [4] Jeffrey Dean and Luiz André Barroso. The tail at scale. *Communications of the ACM*, 56(2):74–80, 2013.
- [5] Herbert Goldstein, Charles P. Poole, and John L. Safko. *Classical Mechanics*. Addison Wesley, San Francisco, 3 edition, 2002.
- [6] Henry M. Levy. *Capability-Based Computer Systems*. Digital Press, Bedford, MA, 1984.
- [7] Alberto G. Rojo and Anthony M. Bloch. The principle of least action. In Robert A. Meyers, editor, *Encyclopedia of Complexity and Systems Science*, pages 5572–5583. Springer, New York, 2009.
- [8] Anton H. Romulus. AHR-v3: A supervisory behavioral engine above the nanokernel. Available at <https://sananton.substack.com/p/manufacturers-defaults-vs-the-virtual>, October 2026. Working paper, AHR #timemachine.
- [9] Edwin F. Taylor. Principle of least action. Scholarpedia, 2011.